Pavan Sai Kumar Reddy Thummala

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.48
Narayana junior collage	Class XII (Telangana state board of intermediate education)	2019	93.40%
Bhashyam public school K.P.H.B Colony, Medchal District	Class X (Board of Secondary Education Telangana State)	2017	9.00%

POSITIONS OF RESPONSIBILITY

Hostel Representative | **Hostel Sports Coordinator** | **DSA Mentor** | **Robotics Mentor**

EXPERIENCE

Junior Research Assistant Fellow

Researcher specializing in the development of autonomous flight systems and heavy-payload multicopter architectures.

- Engineered flight-control systems for heavy-payload UAVs over a 5-month tenure, deploying a Jetson Orin Nano autonomy stack to achieve 30-minute flight endurance with a 100kg payload capacity.
- Developed a ROS-based stereo Visual-Inertial Odometry and path-planning pipeline for GPS-denied environments, validating 100% obstacle-avoidance success at 7 m/s with under 8% trajectory deviation.

Research Intern

Researcher focused on deploying real-time autonomous obstacle avoidance and local navigation systems for outdoor UAV operations.

- Deployed a real-time obstacle avoidance pipeline by implementing the 3DVFH+ algorithm on an NVIDIA Jetson Nano with Intel RealSense D435i depth data, enabling dynamic collision-free trajectory generation.
- Validated the autonomous navigation architecture by optimizing the perception pipeline for edge hardware and executing 20+ outdoor flight tests to ensure reliable collision-free operations.

PROJECTS

Kala - Multicopter UAV Firmware Development

A customized ArduPilot-based C++ firmware designed for secure, high-performance multicopter UAV operations.

- Extended the ArduPilot C++ flight stack by integrating custom tamper-proofing software algorithms, successfully securing the multicopter control architecture against unauthorized modifications for payloads up to 5 kg.
- Validated firmware by executing real-world mission profiling, achieving stable autonomous flight across a 10 km operational range with a 30-minute endurance at 15 m/s.

TihanFly Ground Control Station

A cross-platform ground control station software for UAV command, mission editing, and multi-drone communication

- Engineered a ground control station using C++, Python, Qt, and Boost-ASIO to implement MAVLink-based telemetry routing, successfully orchestrating simultaneous mission command and control for 3 UAVs..
- Validated 6 distinct UAV weight categories and frame structures, achieving reliable cross-platform flight operations.

Multi-Spectral Satellite Image Semantic Segmentation

A deep-learning semantic segmentation pipeline designed to classify target regions from 16-channel satellite imagery.

- Extracted pixel-wise segmentation masks from a 4.09 GB 16-channel multispectral dataset by engineering a PyTorch U-Net utilizing GPU-accelerated AMP.
- Achieved a Dice Coefficient of 0.24244 on the Kaggle evaluation, following 100 epochs of optimization using BCE + Dice Loss and an 85:15 training-validation split.

AI-Driven 5G-V2X Resource Scheduler

A Deep Learning-driven 5G radio resource block scheduler balancing spectral efficiency and network fairness for v2x communications.

- Architected a PyTorch Deep Autoencoder and K-Means pipeline to compress 8 RF parameters into 3 latent clusters, minimizing reconstruction error to a 0.1426 MSE validation loss.
- Achieved 0.839 Jain's Fairness Index (outperforming Max C/I) and 0.112 Mbps average throughput via Pareto-optimal allocation with dynamic speed penalties.

PUBLICATIONS

- TihanFly: A Multi-Drone Ground Control Station for Scalable UAV Operations. | IEEE APSCON.
- Ground Control Station for managing airspace operations in Advanced Air Mobility (Accepted) | VTC2026.
- UAV Based Precise Multi Payload Dropping System Using Vision Guidance and RTK Positioning. | IEEE APSCON.
- Secure RSA Based Text File Encryption and Decryption Using LabVIEW | *Engineering, Science & Sustainability*.
- Deep Learning based Real-time Stolen Vehicle Detection Model with Improved Precision and Reduced Look Up Time.| IEEE

SKILLS

C, C++, Verilog, Python, MATLAB | ML/DL, Computer Vision, Image Processing, Semantic Segmentation, Pytorch | ArduPilot, MAVLink, ROS, n8n